

Proof of Completeness of Commitments

A trivial solution as the first step

Note that the whole construction is based on the assumption that we can extend HIBE to M-HIBE, where the IDs of each dimension can be extended independently.

At high level, when generalizing the commitments (indeed they are the decryption keys of HIBE) of Esha's work from one-dimensional range queries to two-dimensional range queries, we should deal with both dimensions for every commitment. We assume that for each node x in the tree $\mathcal{T}_l$ (full binary tree with $l+1$ levels), there is another tree $\mathcal{T}_l$ associated with the node x , where x is a node that can be represented by a binary string, as defined in Esha's work. To distinguish two types of $\mathcal{T}_l$, we denote the first $\mathcal{T}_l$ as $1st - \mathcal{T}_l$, and the latter one as $2nd - \mathcal{T}_l(x)$ which associates with the node x of $1st - \mathcal{T}_l$. The root node in $\mathcal{T}_l$ is denoted as $\perp$.

Denote $x|y$ as the region (i.e. rectangle) $\underbrace{[x0 \cdots 0, x1 \cdots 1]}_{l\text{-bit}} \times \underbrace{[y0 \cdots 0, y1 \cdots 1]}_{l\text{-bit}}$. Denote each datum in the dataset $\mathcal{D}$ as a point $P(k_\alpha, k_\beta)$. k_α and k_β are **binary strings** representing $1st$ and $2nd$ coordinates of the point, respectively. Each point P 's $1st$ coordinate will be a non-empty leaf node in $1st - \mathcal{T}_l$. If a point P 's $1st$ coordinate has the prefix x , then its $2nd$ coordinate will be a non-empty leaf node in the tree $2nd - \mathcal{T}_l(x)$. $2nd - \mathcal{T}_l(x)$ indeed collects information about non-empty second coordinates of points in $\mathcal{D}$ that show up in $x|\perp$. For example, if $l = 4$, and we have three points $P_1(0010, 0000), P_2(0011, 0010), P_3(0001, 0011)$, P_1 and P_2 's $2nd$ coordinates (i.e. 0000 and 0010) will be non-empty leaf nodes in the tree $2nd - \mathcal{T}_l(001)$, since P_1, P_2 's first coordinate has the prefix 001. The root nodes of $1st - \mathcal{T}_l$ and $2nd - \mathcal{T}_l(x)$ are $\perp$ which represents the prefix for any binary string. When a commitment $commit(x|y)$ is generated, it means the region is empty. The binary string before $|$ denotes the commitment for the $1st$ dimension, and the binary string after $|$ denotes the commitment for the $2nd$ dimension.

The trivial solution is making commitments for all empty regions in each region $\underbrace{[x0 \cdots 0, x1 \cdots 1]}_{l\text{-bit}} \times \underbrace{[0^l, 2^l - 1]}_{l\text{-bit}}$, for all x in $1st - \mathcal{T}_l$. This process can be seen as collecting information of empty regions in each rectangle $x|\perp$ where $x \in \mathcal{T}_l$. It can be interpreted as follows. For all points that fall within $x|\perp$, **ignore their first coordinates**, just focus on what second coordinates show up in that region. We find the empty ranges of second coordinates based on the points in $\mathcal{D}$ falling within that region.

Technically, for each node x in $1st - \mathcal{T}_l$, let $C(x) = \{c_1(x), \dots, c_i(x), \dots, c_s(x)\} = GetRoots(l, [0^l, 2^l - 1], \mathcal{K}(x)_{[0^l, 2^l - 1]})$ (the algorithm *GetRoots* is the same as Esha's, which outputs multiple $root_i$ (denoted as $c_i(x)$ in our work) for the union of canonical coverings of empty ranges), where $\mathcal{K}(x)_{[0^l, 2^l - 1]}$ is the set of coordinates which are represented by non-empty leaf nodes in the $2nd - \mathcal{T}_l(x)$. Then we make commitments for all $i \in [1, s]$, where s is the number of nodes in $C(x)$, and the format of the commitments is $commit(x|c_i(x))$. $x|c_i(x)$ denotes the region $\underbrace{[x0 \cdots 0, x1 \cdots 1]}_{l\text{-bit}} \times \underbrace{[c_i(x)0 \cdots 0, c_i(x)1 \cdots 1]}_{l\text{-bit}}$, and a commitment $commit(x|c_i(x))$ denotes that the region is empty. If the $2nd - \mathcal{T}_l(x)$ is empty, i.e. $C(x) = \{\perp\}$, then the commitment would be $commit(x|\perp)$. $x|bot$ denotes the region $\underbrace{[x0 \cdots 0, x1 \cdots 1]}_{l\text{-bit}} \times \underbrace{[0^l, 2^l - 1]}_{l\text{-bit}}$.

By those commitments made above, any challenge of proof $x|y$ will be verified as true if the region represented by $x|y$ is empty, since we have made commitments for empty regions inside the

region $\underbrace{[x0 \cdots 0, x1 \cdots 1]}_{l\text{-bit}} \times \underbrace{[0^l, 2^l - 1]}_{l\text{-bit}}$ for all $x \in 1st - \mathcal{T}_l$. $x|y$ must be a subregion of $x|y'$ where $y' \in C(x)$ and y' is the prefix of y . Thus, $commit(x|y)$ will be derived by $commit(x|y')$. If the region $\underbrace{[x0 \cdots 0, x1 \cdots 1]}_{l\text{-bit}} \times \underbrace{[y0 \cdots 0, y1 \cdots 1]}_{l\text{-bit}}$ is not empty, the result of verification cannot be true, since there must be at least one point in $x|y$ whose second coordinate lies within $\underbrace{[y0 \cdots 0, y1 \cdots 1]}_{l\text{-bit}}$.

There will not be a $commit(x|y')$ where y' is a prefix of y due to this interception, just as in the one dimensional case.

If we do the above procedures trivially, the size of commitments will be at least 2^l since we make commitments for each node in $1st - \mathcal{T}_l$. Then, in the following approach, we reduce the size of commitments to $O(nl^2)$, by the observation that for all $x \in 1st - \mathcal{T}_l$ which is a suffix of $z \in 1st - \mathcal{T}_l$, the set $\{commit(x|\perp)\}$ s can be reduced to one commitment $commit(z|\perp)$.

Description of the algorithm

Preliminaries

Let $\mathcal{D}$ denote the database. Each datum in $\mathcal{D}$ is denoted as (k_α, k_β, v) , where k_α, k_β are two keys used for range queries and v is a vector representing other values associated with this datum. We ignore v since it is not directly related to the 2D range query. Thus, each datum now is considered as a point. Let P denote a point, and $P_i(k_{\alpha_i}, k_{\beta_i}) \in \mathcal{D}$ denote the point whose first coordinate is k_{α_i} and the second coordinate is $k_{\beta_i}, i \in [1, n]$, where n is the total number of points in $\mathcal{D}$. For each dimension, we use an l -bit binary string to represent the corresponding coordinate, thus a point will be represented by a $2l$ -bit binary string. In each dimension, the range is $[0^l, 2^l - 1]$. Assume the points in $\mathcal{D}$ are sorted in the ascendant order first by $1st$ -dimension coordinate and then by $2nd$ -dimension coordinate. In other words, we have $P_i < P_j$ if $k_{\alpha_i} < k_{\alpha_j}$, or $(k_{\alpha_i} = k_{\alpha_j}) \cap (k_{\beta_i} < k_{\beta_j})$.

Let $x, y, z, a, b, k_{\alpha_i}, k_{\beta_i}$ be binary strings to represent corresponding values from now on.

Let $pref(x)$ be the set of nodes in $\mathcal{T}_l$ such that $z \in pref(x)$ if and only if z is a prefix of x . Let $PR(x)$ be the set of points such that $PR(x) = \{P_i(k_{\alpha_i}, k_{\beta_i}) \in \mathcal{D} | x \in pref(k_{\alpha_i}), i \in [1, n]\}$. Visually, $PR(x)$ means the set of points in $\mathcal{D}$ who fall within $x|\perp$ since their first coordinates have the prefix x . We redefine the method $GetRoots(l, [a, b], \mathcal{K}_{[a, b]})$ of Esha's work. First, we redefine $\mathcal{K}_{[a, b]}$.

Let $\mathcal{K}(I, x)$ be a function which takes an indicator I which indicates the dimension and a binary string x . $I = 1$ denotes the $1st$ dimension and $I = 2$ denotes the $2nd$ dimension. If $I = 1$, we ignore x . $\mathcal{K}(1, x)$ actually denotes the set of $1st$ coordinates $\{k_{\alpha_i}\}_{i \in [1, n]}$ of points in $\mathcal{D}$, i.e. $\mathcal{K}(1, \perp)$ is the set of all non-empty leaf nodes in $1st - \mathcal{T}_l$. If $I = 2$, the function takes $PR(x)$ and the output is a set of $2nd$ coordinates $\{k_{\beta_i}\}$ of points in $PR(x)$. In fact, $\mathcal{K}(2, x)$ represents all non-empty leaf nodes in $2nd - \mathcal{T}_l(x)$. Then $\mathcal{K}(I, x)_{[a, b]}$ means a subset of $\mathcal{K}(I, x)$ in which the coordinates also fall within $[a, b]$.

We now have $GetRoots(l, [a, b], \mathcal{K}(I, x)_{[a, b]})$ which runs the same as $GetRoots(l, [a, b], \mathcal{K}_{[a, b]})$. It returns a set of nodes that represents the union of the canonical coverings of the empty ranges according to the set of coordinates in $\mathcal{K}(I, x)_{[a, b]}$.

0.1 The process to generate commitments

0.1.1 Data owner side

We define two kinds of commitments as follows, based on the dimension we are focusing on.

- **Step 1.** We focus on the first dimension commitment, where the second dimension commitment is fixed to $\perp$ (the set of commitments is denoted as $Commit(*|\perp)$). In this phase, we find all empty regions of the type $*|\perp$ which means no point lie within $*|\perp$. Let $R = \{c_1, \dots, c_i, \dots, c_s\}$ be the result returned by $GetRoots(l, [0^l, 2^l - 1], \mathcal{K}(1, \perp)_{[0^l, 2^l - 1]})$.

$$Commit(*|\perp) = \{commit(c_1|\perp), \dots, commit(c_i|\perp), \dots, commit(c_s|\perp)\},$$

where s is the number of nodes in R . Intuitively, each commitment $commit(c_i|\perp)$ indicates that the region $\underbrace{[c_i 0 \dots 0, c_i 1 \dots 1]}_{l\text{-bit}} \times [0^l, 2^l - 1]$ is empty. Note that if there is no point in $\mathcal{D}$, $commit(\perp|\perp) \in Commit(*|\perp)$.

- **Step 2.** We focus on the second dimension commitment, where the first dimension commitment is fixed to be x where $x \in 1st - \mathcal{T}_l$, and the commitment set is denoted as $Commit(x|*)$. Let $Q(x)$ denote the result returned by $GetRoots(l, [0^l, 2^l - 1], \mathcal{K}(2, x)_{[0^l, 2^l - 1]})$. **If $Q(x)$ is $\{\perp\}$, we abort since $x|\perp$ is generated/can be derived in the previous phase.** This is easy to understand since it means no point lie within $x|\perp$.

For x of which $Q(x)$ is not $\{\perp\}$,

$$Commit(x|*) = \{commit(x|d_1), \dots, commit(x|d_i), \dots, commit(x|d_t)\},$$

where d_i is a binary string representing a node in $Q(x)$, and t is the number of nodes in $Q(x)$. For example, if $l = 4$, and we have three points $P_1(0010, 0000), P_2(0011, 0010), P_3(0001, 0011)$. For $x = 001$,

$$Commit(001|*) = \{commit(001|0001), commit(001|0011), commit(001|01), commit(001|1)\},$$

where the 2nd-dimension commitment $\{0001, 0011, 01, 1\}$ is the union of canonical coverings of empty ranges in $[0000, 1111]$, since $PR(001) = \{P_1, P_2\}$ and 2nd-dimension coordinates of P_1 and P_2 are (0000 and 0010). Note that x can be $\perp$. **Note that $commit(x|\perp) \notin Commit(x|*)$ since we abort if $Q(x)$ is $\perp$.**

In fact, until now, we have described the whole story of the algorithm to generate commitments. That is, first take all first coordinates and generate $Commit(*|\perp)$. Then for any $x \in \mathcal{T}_l$, generate $Commit(x|*)$. Since if there is no point lie within $x|\perp$, it is not necessary to generate $Commit(x|*)$, we can skip the commitment generation for those x s. Indeed, we can predict which x we should skip before Step 2 by the result of Step 1. That is, if x is a suffix of nodes in R (returned by $GetRoots(l, [0^l, 2^l - 1], \mathcal{K}(1, \perp)_{[0^l, 2^l - 1]})$), obviously this x should be skipped. Indeed, another interpretation is that we only need to care x where there are some points whose first coordinates have the prefix x .

By the above description, we can also see the commitments generated by the algorithm is a more efficient version of the commitments of the trivial solution.

0.1.2 Client side

However, we just described how the data owner generates commitment according to their knowledge of the dataset. Now we describe how the client generate challenges that they wants the server to prove. The challenge means the empty regions that the client sees according to the answers they get and the range they query.

Assume the queried range is $[a, b] \times [c, d]$ and assume the canonical covering of $[a, b]$ is $\{u_1, \dots, u_s\}$ and the canonical covering of $[c, d]$ is $\{v_1, \dots, v_t\}$. We define the 2D variant of the $GetRoots(l, [a, b], \mathcal{K}(I, x)_{[a, b]})$ algorithm, that is $GetRoots(l, [a, b] \times [c, d], \mathcal{K}(I, x)_{[a, b] \times [c, d]})$. Based on the definition we have for $\mathcal{K}(I, x)$, $\mathcal{K}(I, x)_{[a, b]}$ is just a truncated version where the range in our vision is truncated to $[a, b] \times [c, d]$. That means, $\mathcal{K}(1, x)_{[a, b] \times [c, d]}$ is the first coordinates of set of points in $\mathcal{D}'$ (i.e. the answer). $Q = GetRoots(l, [a, b] \times [c, d], \mathcal{K}(1, x)_{[a, b] \times [c, d]})$ will return the union of the canonical coverings of empty ranges in $[a, b]$ based on $\mathcal{K}(1, x)_{[a, b] \times [c, d]}$. The client will challenge on $c_i|v_j$ (we will explain in the next section why this is the case) for all $c_i \in Q, j \in [1, t]$. For each point P'_i in $\mathcal{D}'$, we denote the first coordinate of the point as x'_i , $\mathcal{K}(2, x'_i)_{[a, b] \times [c, d]}$ is set of the second coordinates of points in $\mathcal{D}'$ falling with $x'_i|c, d]$. $Q'(x_i) = GetRoots(l, [a, b] \times [c, d], \mathcal{K}(2, x'_i)_{[a, b] \times [c, d]})$ will return the union of the canonical coverings of empty ranges in $[c, d]$ based on $\mathcal{K}(2, x'_i)_{[a, b] \times [c, d]}$. For each x'_i , The client will challenge on $x'_i|d_j$ for all $d_j \in Q'(x'_i)$. Note that x_i indeed are non-empty leaf nodes in $\mathcal{T}_l$ in the range $[a, b]$.

How to prove a region is empty

Assume the client asks to prove a region $[x_{lo}, x_{hi}] \times [y_{lo}, y_{hi}]$ is empty, where $x_{lo}, x_{hi}, y_{lo}, y_{hi}$ are l -bit binary strings representing the corresponding coordinate.

Assume $\{c_1, c_2, c_3\}$ is the canonical covering of the range $[x_{lo}, x_{hi}]$, where c_i is a binary string, $i \in [1, 3]$ and $\{d_1, d_2, d_3\}$ is the canonical covering of the range $[y_{lo}, y_{hi}]$, where d_i is a binary string, $i \in [1, 3]$, the query for the proof should be $\{c_1, c_2, c_3\} \times \{d_1, d_2, d_3\}$, i.e. $c_1|d_1, c_1|d_2, c_1|d_3, c_2|d_1, c_2|d_2, c_2|d_3, c_3|d_1, c_3|d_2, c_3|d_3$.

In general, we make the following statement.

Statement: Given a region $[x_{lo}, x_{hi}] \times [y_{lo}, y_{hi}]$, and $C = \{c_1, \dots, c_i, \dots, c_s\}$ is the canonical covering of the range $[x_{lo}, x_{hi}]$, $D = \{d_1, \dots, d_j, \dots, d_t\}$ is the canonical covering of the range $[y_{lo}, y_{hi}]$, then the region can be represented by the union of $\{c_i|d_j\}$ for all $c_i \in C, d_j \in D, i \in [1, s], j \in [1, t]$. s, t are the number of nodes in the corresponding set, respectively.

Proof: We can split the region by 1st-dimension coordinate and 2nd-dimension coordinate. Firstly, we split the region by 1st-dimension coordinate. That is, the range of the 1st-dimension $[x_{lo}, x_{hi}]$ can be represented by the union of $\{c_i\}$ where $c_i \in C, i \in [1, s]$. Then, the region can be represented by the union of the subregions $\underbrace{[c_i 0 \dots 0, c_i 1 \dots 1]}_{l\text{-bit}} \times [y_{lo}, y_{hi}], i \in [1, s]$. Then, we

split each subregion by 2nd dimension coordinate. The range $[y_{lo}, y_{hi}]$ can be represented by $D = \{d_1, \dots, d_j, \dots, d_t\}$. Thus, for each subregion, it can be divided to subsubregions $[c_i 0 \dots 0, c_i 1 \dots 1] \times [d_j 0 \dots 0, d_j 1 \dots 1]$, which is $c_i|d_j$ (where $i \in [1, s], j \in [1, t]$). Thus, we complete the proof.

To ask for a proof that the region $[x_{lo}, x_{hi}] \times [y_{lo}, y_{hi}]$ is empty, first the client will generate canonical coverings of the ranges in each dimension, denoted as C and D , for each dimension respectively. Then, the client combines each node $c_i \in C$ with each canonical node $d_j \in D$ to form the subregion $c_i|d_j$. The client asks for proofs for each subregion. The size of proof for each rectangle will be $O(l^2)$.

Recall that in the previous section, we let the client challenge on $c_i|v_j$ for all $c_i \in Q, j \in [1, t]$. They are indeed the canonical coverings of the rectangles $c_i|[c, d]$.

Proof of Completeness of Commitments

The completeness of commitments means, for any query for a proof of $p|q$, where p, q are binary strings, if the region represented by $p|q$ is empty, we can find a commitment $commit(p'|q')$

to generate a proof for $p|q$, with 100% certainty. p', q' is the prefix of p, q respectively. Note that $\perp$ is the prefix for any string.

We prove it by two steps. We assume the whole region $[0^l, 2^l - 1] \times [0^l, 2^l - 1]$ is not empty, which means there is no $\text{commit}(\perp|\perp)$. If the whole region is empty, a $\text{commit}(\perp|\perp)$ is sufficient to generate all proofs.

Step 1

In this step, we prove that if we generate commitments on **each** node x (the trivial solution) of the full binary tree $1st - \mathcal{T}_l$, as $\text{Commit}'(x|*)$, we will definitely find a commitment $\text{commit}(p'|q')$.

Statement: For any challenge on $p|q$, where p, q are binary strings, if the region $p|q$ is empty, then there must be a commitment $\text{commit}(p|q') \in \text{Commit}'(p|*)$ that q' is the prefix of q .

Proof: Let $Q(p)$ be the output of $\text{GetRoots}(l, [0^l, 2^l - 1], \mathcal{K}(2, p)_{[0^l, 2^l - 1]})$, denoted as $Q(p) = \{d_1, d_2, \dots, d_i, \dots, d_t\}$ is the canonical coverings of empty ranges based on the second coordinates of points in $\mathcal{D}$. $\text{Commit}'(p|*) = \{\text{commit}(p|d_1), \dots, \text{commit}(p|d_i), \dots, \text{commit}(p|d_t)\}$. By the definition of canonical covering of the range, if the region $p|q$ is empty, then there must be some $d_i = q' \in Q(p)$, which is the prefix of q . That means $\text{commit}(p|q') \in \text{Commit}'(p|*)$.

Step 2

We will now show that the two types of commitments, i.e. $\text{Commit}(x|*)$, and $\text{Commit}(*|\perp)$ is sufficient to generate all $\text{Commit}'(x|*)$ in the first step, for all x in the full binary tree.

The proof is obvious. For $x \in \mathcal{T}_l$ where there are no point in $\mathcal{D}$ falling in $x|\perp$, the commitment $\text{commit}(x|\perp)$ can be derived by a $\text{Commit}(z|\perp)$ where $z \in \text{GetRoots}(l, [0^l, 2^l - 1], \mathcal{K}(1, x)_{[0^l, 2^l - 1]})$, since z is a node in the canonical covering of the empty ranges for first coordinates. Thus, we can use $\text{Commit}(*|\perp)$ type commitments to cover $\text{commit}(x|\perp)$ where there are no point in $\mathcal{D}$ falling in $x|\perp$. For $x \in \mathcal{T}_l$ where there are at least one point falling within $x|\perp$, the $\text{Commit}(x|*)$ type commitments which are exactly the same as the commitments in the trivial solution. Thus, the two types of commitments $\text{Commit}(x|*)$, and $\text{Commit}(*|\perp)$ are sufficient to generate commitments as in the trivial solution.

Complexity Analysis

There are two types of commitments. $\text{Commit}(*|\perp)$ and $\text{Commit}(x|*)$.

The number of commitments in $\text{Commit}(*|\perp)$ is $O(nl)$, since there are $O(n)$ different empty ranges and each range should be represented with $O(l)$ nodes.

Now let us compute the number of commitments in $\text{Commit}(x|*)$ with respect to x . If there are m points falling within $x|\perp$, the number of nodes in $\text{GetRoots}(l, [0^l, 2^l - 1], \mathcal{K}(2, x)_{[0^l, 2^l - 1]})$ are $O(ml)$. For each level of the tree $\mathcal{T}_l$, e.g. the i th level, for each node y in that level, $|\text{Commit}(y|*)| = O(m_y l)$ where m_y means the number of points falling within $y|\perp$. In total, for all y in the i th layer, the sum of commitments $\text{Commit}(y|*)_{y \text{ in } i\text{th level}}$ is $O(nl)$ since each point can only show up in one region once. That holds for each layer. In total, there are $l + 1$ layers, thus, the sum of $\text{Commit}(x|*)$ type commitments are $O(nl^2)$.

In total, the complexity for storage of commitments is $O(nl^2)$

Similarly, from the perspective of the client, the number of challenge they need to issue is $O(m_{\mathcal{D}} l^2)$ where $m_{\mathcal{D}}$ is the number of points in the answer, since they find $O(m_{\mathcal{D}})$ rectangles and each rectangles can be represented by $O(l^2)$.

1 Generalize to $d \geq 2$

By the same principle, for example when $d = 3$, we can first only consider the first coordinates generate $Commit(*|\perp|\perp)$, and then fix the first coordinate to generate two types of commitments as in 2D case.